ARYAN MEENA

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EDUCATION

Boston University, Metropolitan College — Boston, MA
*M.S., Applied Data Analytics — **GPA: 3.75 / 4.0*** — *Jan 2025 – May 2026*

- **Relevant Coursework:** Advanced Machine Learning, Foundations of Machine Learning, Data Mining, Big Data Analytics, Data Science with Python, Data Visualization, Web Mining & Graph Analytics, Analysis of Algorithms

Indian Institute of Space Science and Technology (IIST) — Thiruvananthapuram, India
B.Tech., Electronics and Communication Engineering — *Aug 2019 – May 2023*

- **Relevant Coursework:** Computer Vision, Deep Learning, Machine Learning for Signal Processing, Clustering, Regression, Expectation Maximization

PROFESSIONAL EXPERIENCE

Research Assistant — Quantitative FX & DXY Strategy Research — Boston, MA
Boston University, Prof. Eugene Pinsky (Short Paper – Traders Magazine) — *Sept 2025 – May 2026*

- Designed and backtested **24 intraday sub-period trading strategies** across the DXY U.S. Dollar Index and 6 major FX pairs (EUR, GBP, JPY, CAD, CHF, SEK) on **15 years of daily OHLC data** (2010–2025), decomposing each trading day into overnight (close-to-open) and daytime (open-to-close) sessions with position rules: Long, Short, Cash, Inertia (momentum), and Reversal.
- Best DXY strategy (Short overnight / Long daytime) achieved a **final balance of $318.80 from $100** over the 15-year backtest — a **+219% cumulative return** vs. **+22% for buy-and-hold**; **Sharpe ratio 1.06** vs. 0.22 B&H; max drawdown only **–13.1%** vs. –15.3% B&H.
- Rigorously evaluated strategy robustness under 0, 1, and 2 basis-point transaction cost regimes; produced styled performance tables (Excel + Python) and authored a **short paper accepted by *Traders Magazine* (est. 1982)** reporting intraday FX return decomposition findings.
- *Stack: Python, pandas, NumPy, openpyxl, Matplotlib; 15-year FX OHLC dataset (7 instruments)*

Teaching Assistant — Advanced Machine Learning & Data Science with Python — Boston, MA
Boston University, Metropolitan College — *Jan 2026 – May 2026*

- Supported 60+ graduate students in MET CS 767 (Advanced ML) and MET CS 577 (Data Science with Python) with model implementation, debugging, evaluation metrics, and theoretical ML concepts.
- Held weekly office hours, graded assignments, and provided feedback on course projects involving regression, classification, clustering, and deep learning.

Machine Learning Research Intern — Remote / Bangalore, India
Indian Space Research Organisation (ISRO) — *Jun 2022 – Aug 2022*

- Built a transfer-learning CNN pipeline (VGG-16 backbone) for automated road detection in multi-spectral satellite imagery, achieving **82.03% accuracy** on raw (non-curated) images.
- Validated model robustness across heterogeneous terrain types; delivered a deployable inference pipeline reducing manual annotation time by an estimated **60%**.

AWARDS & HONORS

- **3rd Place — MedAI Hackathon 2026** — *Boston University*
  Built a 3D MedicalNet ResNet-34 with FiLM tracer conditioning to predict amyloid PET Centiloid scores across 4 tracers (FBB, FBP, NAV, PIB); validation MAE **7.31 CL**, Pearson r **0.968**; leaderboard test MAE **12.67 CL** — 3rd of 40+ teams.
- **Teaching Assistant Award (Merit Selection)** — Boston University, Spring 2026
- **Dean's List** (GPA 3.75/4.0) — Boston University, 2025–2026

PUBLICATIONS & SUBMITTED WORK

- **Meena, A.** et al. "Intraday Sub-Period Return Decomposition Strategies on DXY and FX Markets." *Traders Magazine (est. 1982)*, 2026. Backtested 24 strategies over 15 years (2010–2025); best strategy achieved **Sharpe 1.06**, **+219% cumulative return** vs. +22% buy-and-hold on DXY; robustness-tested under 0–2 bps transaction costs.

- **Meena, A.** et al. "Accelerated CNN Training with Genetic Algorithm Optimization." *IEEE IATMSI*, 2024. Reduced CNN training time by $>$**50%**; proposed *DoubledMNIST*, a harder benchmark for realistic model evaluation.
- **Meena, A.** et al. "Enhanced Scene Text Recognition for Accessibility." *ICADIE (Springer)*, 2024. Multi-stage CV pipeline (SVM + NMS + language model) expanded vocabulary from 50 to 427 words, improving recognition accuracy by **4%**.

MAJOR PROJECTS

WikiFlow: Distributed Wikipedia Editor Retention Prediction *github.com/RyanSingh0/wikiflow*

- Built an end-to-end **distributed ML pipeline on Google Cloud Platform (GCP)** processing **25 GB** of Wikipedia revision history (**1.16M editors**, 60M+ revision events) using PySpark on Dataproc.
- Engineered 13 behavioral features from each editor's first 10 edits; trained three models from scratch on PySpark RDDs: custom logistic regression (**AUC 0.845**), K-Means clustering (silhouette 0.335), and an Elephas distributed Keras DNN (**AUC 0.909**).
- Published interactive Plotly dashboard to GCS and results to BigQuery; full pipeline runs end-to-end in under 2 hours with zero local data movement.
- *Stack: PySpark, GCP (Dataproc, GCS, BigQuery), TensorFlow/Keras, Elephas, Plotly*

Driver Drowsiness Detection System *github.com/RyanSingh0/drowsiness-detection*

- Designed a modular **4-stage computer vision + temporal ML pipeline**: face detection (RetinaFace/FPN), facial keypoint regression (ResNet-18 with confidence-aware loss), geometric feature extraction (EAR/MAR), and temporal classification.
- Achieved **ROC-AUC 0.964** and **F1-score 0.923** on the NTHU-DDD driving dataset; selected Random Forest over LSTM due to superior test-set generalization (F1: 0.923 vs. 0.752 for logistic regression).
- Novel contribution: confidence-aware landmark regression loss function that explicitly trains the model to estimate its own uncertainty.
- *Stack: PyTorch, OpenCV, RetinaFace, scikit-learn, NumPy*

MedAI Hackathon: Amyloid PET Centiloid Prediction (3$^{\text{rd}}$ Place) *github.com/RyanSingh0/medai-amyloid*

- Built a **3D MedicalNet ResNet-34 with FiLM (Feature-wise Linear Modulation) tracer conditioning** to predict amyloid burden (Centiloid scale) from volumetric PET brain scans across 4 tracers (FBB, FBP, NAV, PIB) in a single unified model.
- Achieved validation **MAE 7.31 CL, Pearson r = 0.968** on 500-sample held-out set; **test MAE 12.67 CL on unseen leaderboard data — 3$^{\text{rd}}$ of 40+ teams**.
- Training: AdamW with differential LRs (backbone lr/10), Huber loss (δ=15), mixed precision (torch.amp), cosine LR annealing, gradient clipping, early stopping — 40 epochs.
- *Stack: PyTorch (3D Conv), MedicalNet pre-trained weights, FiLM conditioning, Huber loss*

Learning Difficulty Screening — NHIS 2024 Classification Pipeline *github.com/RyanSingh0/nhis-learning-difficulty*

- Built an end-to-end classification pipeline on 7,439 children from the 2024 National Health Interview Survey (353 features) to screen for childhood learning difficulties.
- Trained **36 models** across 9 algorithms (LR, DT, NB, k-NN, SVM-RBF, MLP, RF, XGBoost, Bagging) × 4 balancing strategies (downsample, upsample, SMOTE, NearMiss-1); best model: **XGBoost + downsampling**, exceeding minimum TPR constraints.
- *Stack: R (caret, xgboost, ROSE), Python, ggplot2*

Hospital Readmission Prediction (Healthcare Analytics) *github.com/RyanSingh0/hospital-readmission*

- Analyzed **30,000 patient discharge records** (7.17:1 class imbalance) to predict 30-day hospital readmissions; engineered 5 domain-driven features and tuned decision thresholds.
- Final Random Forest + SMOTE model achieves **ROC-AUC 0.5654**, capturing **35% of readmissions**; SHAP analysis identified top predictors; projected **$41.35M net savings (ROI 2,761%)** per hospital system.
- *Stack: Python, scikit-learn, SHAP, SMOTE (imbalanced-learn), Matplotlib, Seaborn*

Predictive Maintenance on Jet Engine Sensors (NASA C-MAPSS) *github.com/RyanSingh0/jet-engine-rul*

- Modeled run-to-failure time-series data for 100 turbofan engines; identified statistically significant degradation sensors via t-tests and ANOVA ($p < 0.001$).
- Multiple linear regression (Sensor11 + Sensor9) achieved $\mathbf{R^2 \approx 0.50}$; regularized logistic regression (Lasso/Elastic Net) for failure classification reached **AUC 0.945** and **96.52% accuracy** on the test set.

- *Stack: R, ggplot2, caret, ANOVA, ANCOVA, Lasso/Ridge/ElasticNet*

MBTA Subway Network Community Detection *github.com/RyanSingh0/mbta-community-detection*

- Modeled Boston's subway as a weighted directed graph (106 nodes, 113 edges) from GTFS data; applied Louvain, METIS, and Spectral Clustering for community detection.
- Louvain achieved **modularity 0.8113** with 10 balanced communities; time complexity validated as O(n log n) across subgraph sizes.
- *Stack: Python, NetworkX, python-louvain, METIS, NumPy, Matplotlib*

Wireless V2X Network Performance Analysis *github.com/RyanSingh0/v2x-network-analysis*

- Analyzed TiHAN-V2X dataset (13,208 observations, V2V + V2I scenarios) using ANOVA, chi-square tests, and polynomial regression; key finding: scenario configuration drives throughput far more than signal quality.
- Final multiple regression model explains **86.9% of throughput variance** (Adjusted $R^2 = 0.869$); decision tree classifier achieved 64% accuracy on held-out set.
- *Stack: R, Python, ggplot2, regression, ANOVA, decision trees*

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, C++
ML / DL Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, Gradient Boosting, Random Forest, CNNs, RNNs/LSTMs
Big Data & Cloud: Apache Spark (PySpark), Google Cloud Platform (Dataproc, GCS, BigQuery), Hadoop, Elephas
Data Science & Analytics: Pandas, NumPy, SHAP, SMOTE, EDA, Feature Engineering, A/B Testing, Hypothesis Testing
Statistics: Regression (Linear, Logistic, Regularized), ANOVA, ANCOVA, Chi-square, Time-Series Analysis
Computer Vision: OpenCV, RetinaFace, ResNet, VGG, Transfer Learning, CNNs
Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau
Tools & DevOps: Git/GitHub, Linux, Jupyter, VS Code, Docker (basic), LaTeX